

While purchasing a new washing machine in an online store, you are recommended to buy connection pipes and detergents. This is an example of which of the following tasks?

Subgroup Discovery

Association rule mining

Descriptive clustering

Predictive clustering

Regression

Which statement(s) are true about grading models?

Linear regression is an example of a grading model.

They create segments within the instance space.

They have a fixed and a finite resolution.

Tree-based models are examples of grading models.

SVM is an example of grading models

Which statement(s) are true about probabilistic models?

Likelihood ratio is the product of posterior odds and prior odds.

Maximum a posteriori is used to face the overfitting problem

In a classification task, the prior probability refers to our prior knowledge about the value of features.

It is preferred to use the likelihood ratio when it cannot be assumed that the prior distribution is uniform.

Posterior odds is the product of likelihood ratio and prior odds.

Sometimes, separate “training, test, and validation” sets are used in cross-validation. What is/are the validation set(s) used for?

To evaluate the final model accuracy after all training and tuning is complete.

For evaluation of the classifications made by the model using numerous performance metrics.

For finetuning hyper-parameters on a separate set than the train/test

To monitor model performance during training and help select the best model.

To make predictions during the inference phase

Which one(s) among the following is considered a property of decision trees?

They require extensive data preprocessing like feature scaling.

They can be applied on both categorical and numerical data

It takes longer to train a decision tree model than to perform inference with it

The basic decision tree algorithm is highly resistant to overfitting.

Select the option(s) that are true about one-versus-one classifiers.

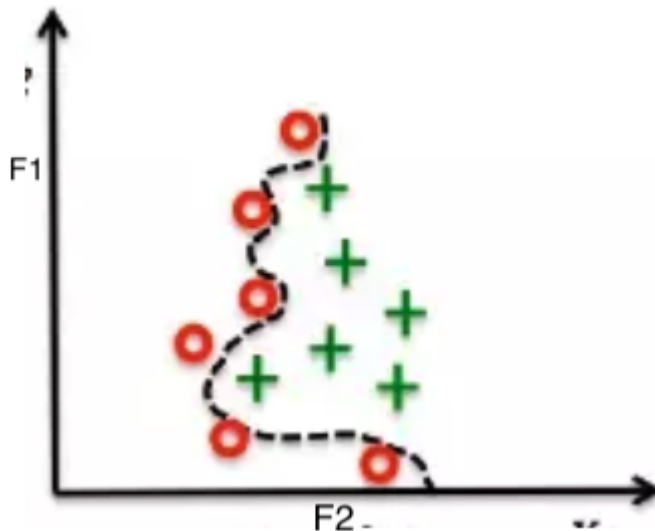
One model is trained per class, distinguishing between that class and all others.

A separate binary classifier is trained for every possible pair of classes.

The “one-versus-one” strategy can be applied in combination with binary classifiers like SVM.

The training data is split so that each binary classifier only sees examples from two classes.

The figure below shows a decision boundary learned by a classifier based on two features (F1 and F2) shown on the two axes. Considering “the bias and variance terms of the error”, the following classifier:



- Has high bias and high variance
- Has high bias and low variance
- Has low bias and low variance
- Has high variance and low bias

Which one(s) of the following shows that the binary classifier clf1 outperforms clf2?

- The R^2 of clf1 is higher than that of clf2.
- The area under the ROC curve of clf1 is lower than that of clf2.
- The RMSE of clf1 is lower than that of clf2.

Imagine the following dataset with only one numerical feature (demonstrated on the x-axis) and two class labels (., +). Which technique(s) can find a decision boundary that can classify this dataset perfectly?



- A decision tree
- Expectation-maximization with two normal distributions (one for “x”, one for “.”).
- A linear margin-maximizing classifier (like the SVM) in combination with a kernel.
- A linear regression model.

Which of the following is/are true for linear models?

- They are stable and not affected by variations in data.
- Linear models are parametric.
- Linear models are prone to overfitting.

The effectiveness of an SVM classifier depends upon (multiple possible):

Selection of kernels

Soft margin complexity hyperparameter C

The model's ability to avoid regularization altogether.

What can principal components analysis (PCA) be used for?

Reducing the dimensionality of the data while preserving as much variance as possible.

Finding the best clusters among data.

Identifying the principal features used by a classification algorithm to perform a classification task.

For feature selection among categorical and numerical features

Which of the following are not an example of an ensemble learning algorithm? (multiple possible)

Random Forest

Adaboost

Gradient Boosting

Decision Trees

Which one of the following is more probable for a right-skewed distribution?

Mode < Median < Mean

Mean = Median = Mode

Mode < Mean < Median

Mean < Median < Mode

After training a linear classifier to make a distinction between two classes based on a single feature, you realize a large amount of error. What is the best strategy to reduce the error without using a different machine learning algorithm?

Stacking

Bagging

Boosting

Sub-space sampling

Neural Networks ____ :

Are trained through an optimization approach on a convex surface formed based on the loss function.

Always output values between 0 and 1

Can be used for regression as well as classification

Can be used in an ensemble

Which one of the following activation functions is suitable for the last layer of a feedforward neural network that is trained for multi-class classification

Linear

Tanh

Softmax

Sigmoid

Which test can be used to compare the performance of multiple algorithms over multiple datasets?

Null hypothesis
Friedman test
T-test
Stratified cross-validation

A company designs a machine learning model to process the CVs of job applicants. Which of the following statements are highly probably to happen?

The model may inherit biases from historical data, leading to discrimination against certain groups of applicants.

The model will always be completely neutral and objective in its evaluation of applicants.

The use of the model can reduce human bias and result in fairer hiring decisions.

The model will always make more accurate decisions than human recruiters.

Explain what you know about 'the curse of dimensionality':

Explain bagging and boosting approaches. What is the difference between these methods in terms of bias and variance?

Suppose a classification problem where one of the possible outcomes is extremely uncommon. Would you prefer to use MAP or MLE in this case? Why?

Here is a simple perturbation-based explainer algorithm (similar to the LIME algorithm, Lecture 11). This algorithm contains the steps to train a surrogate model g for a specific sample x . This sample comes from the initial dataset $X = \{x_1, \dots, x_n\}$.

In a similar fashion to the LIME algorithm, we make use of K-Lasso to train a simple linear surrogate model, and we return the weights w as an explanation to the input features.

Algorithm 1 Explainer algorithm

Require: Classifier f , Number of samples N , Instance $x \in X$, and x' (interpretable version), $X = \{x_1, \dots, x_N\}$

Require: Similarity metric π_x , Length of explanation K

```
1:  $Z \leftarrow \{\}$ 
2: for  $i \in \{1, 2, 3, \dots, N\}$  do
3:    $z'_i \leftarrow \text{sample\_around}(x')$ 
4:    $Z \leftarrow Z \cup (z'_i, f(z_i), \pi_x(z_i))$ 
5: end for
6:  $w \leftarrow K - \text{Lasso}(Z, K)$     {with  $z'_i$  as features,  $f(z)$  as target and  $X$  the input}
7: return  $w$ 
```

With the function `sample_around` to generate new samples in the neighborhood of the interpretable version x' of the original input x

23a

Is there any issue with the proposed Algorithm 1? Explain your answer.

23b

What are the two drawbacks of the original LIME algorithm?

23c

Which of the following statements are valid:

Probing and LIME is part of the local explanations category in xAI

The LIME algorithm does not take into account the complexity of the black-box model.

LIME is a model-specific method

LIME explanations are generated after the training of the black-box model

The LIME algorithm does not take into account the complexity of the surrogate model.

A data scientist in a telecom company is using a decision tree to perform a binary classification by predicting which customers will stop their subscription in the next 6 months.

The first model trained has a very low accuracy on a validation set. Which of the following options will potentially help improving the accuracy of the model on an unseen test set?

Select all that apply.

Hyperparameter tuning.

Normalising the numerical features.

Collecting more labelled data for training.

Reduced error pruning.

A data scientist released his Python code on GitHub that implements the best split for decision trees. You would like to use his code in your project for developing decision trees.

The code is as follows:

```
def _best_split(self, features, labels):
```

```
    """ Determine the best feature to split on.
```

```
        :param features: a pd.DataFrame with named training
        feature columns
```

```
        :param labels: a pd.Series or pd.DataFrame with
        training labels
```

```
        :return: `best_so_far` is a string with the name of
        the best feature,
```

```
        and `best_so_far_loss` is the loss on that feature
```

```

        For each candidate feature the weighted loss of the "yes"
        and "no"
```

```
        instances for that feature are computed
        using self._metric (Impurity metric).
```

```

        We select the feature with the lowest weighted loss.
```

```
    """
```

```
    best_so_far = None
```

```
    best_so_far_loss = 1 # we are only interested
```

```
    #splits that are better than the parent
```

```
    for candidate in features.columns:
```

```
        yes_index = features[candidate] > 0.5
```

```
        yes_labels = labels[yes_index]
```

```

no_labels = labels[~yes_index]
yes_loss = self._metric(yes_labels) #Impurity metric
no_loss = self._metric(no_labels)
candidate_loss = yes_loss
    * (len(yes_labels) / len(labels)) + no_loss
    * (len(no_labels) / len(labels))
if candidate_loss > best_so_far_loss:
    best_so_far = candidate
    best_so_far_loss = candidate_loss
return best_so_far, best_so_far_loss

```

25a

Is it a good idea to use this code to split your decision tree? If not, what is the issue?

25b

Now, you are trying to decide which Impurity function to use. What are (in general) the main differences between entropy and the $\sqrt{\text{Gini}}$ (square root of Gini) index?